

ZOO4307C Lecture 3

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Notes compiled by ZX

What makes a vertebrate a vertebrate?

**The evolution of vertebrates is driven by the switch from filter-feeding to active predation.*

They have to detect prey, process sensory information, orient themselves toward prey, move more rapidly, process the food, supply muscle with oxygen, and finally remove metabolic wastes.

**General trends for these synapomorphies:*

Features of Vertebrates:

1. Genetic & developmental innovations
 - ❖ Hox gene:
 - ❖ Neural crest cells:
 - ❖ Neural placode:
2. Support & locomotion
 - ❖ Mineralized tissue: 2 skeletal materials, cartilage and bone
 - Cartilage:
 - Bone:
 - ❖ Dermal bone vs. endochondral bone
 - Dermal bone:
 - Endochondral bone:
 - ❖ Teeth: enamel, dentine, and cementum
 - Enamel:
 - Dentine:
 - Cementum:

*Enamel and dentine also found in some types of fish scales

- ❖ Keratin:
- ❖ Vertebral element:
- ❖ Fins:
- ❖ W-shaped myomere:

3. Head & brain

Sensory and nervous processing concentrated towards the anterior end.

- ❖ Head extends beyond notochord:
- ❖ Anterior end of nerve tube forms tripartite brain: fore-, mid-, and hind-
 - Forebrain:
 - Midbrain:
 - Hindbrain:
- ❖ Cranium: dermatocranium, chondrocranium, and splanchnocranium
 - Dermatocranium:
 - Chondrocranium:

- Splanchnocranium:
- ❖ Sensory system: vision, mechanoreception, lateral line system, inner ear, otolith organs, hearing, and electroreception
 - Vision:
 - Mechanoreception:
 - Lateral line system:
 - Inner ear:
 - Otolith organs:
 - Hearing:
 - Electroreception:

4. Feeding (digestion) & respiration: better gas exchange and food processing

- ❖ Structural changes in pharynx
- ❖ Pharynx become more muscularized
- ❖ Gut also become more muscularized

5. Physiology: circulation, excretion, and osmoregulation

A more active animal has a higher metabolic demand, muscle needs more ATP (oxygen and nutrients), and produces carbon dioxide and metabolic wastes.

- ❖ Close circulatory system
- ❖ Hemoglobin: transporting oxygen
- ❖ Chambered heart:
- ❖ Fish: single circulation vs. Tetrapods: double circulation

- ❖ Kidney: filtration, ion-regulation, water balance, waste excretion, and osmoregulation
- ❖ Osmoregulation: Body fluid hypo-osmotic relative to seawater
- ❖ Cloaca: urinary + reproductive + digestive common space

